Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

WSQ - CompTIA Certified Data+ Training

TGS Ref No: TGS-2024049212

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
4.0	1 September 2024	Previous release — CompTIA Data+ course slides and lesson plan (legacy master deck).	Dr. Alfred Ang
5.0	19 August 2026	Full rebuild against the CompTIA Data+ (DA0-001) exam domains and the approved Course Proposal (LU1-LU5 / LO1-LO5, TSC ATP-PIN-3001-1.1). 5-day / 40-hour schedule with 15 hands-on labs; assessment block moved to Day 5, 4:00-6:00 pm.	Dr. Alfred Ang

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Course Information

Course Title	WSQ - CompTIA Certified Data+ Training
WSQ Course Reference	TGS-2024049212
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 8 training hours per day (40 hours) — 19 h classroom facilitation + 19 h practical + 2 h assessment = 40 hours
Daily Timing	9:00 am – 6:00 pm (1-hour lunch; tea breaks within training time)
Certification Alignment	CompTIA Data+ (DA0-001) — all five exam domains at their examined weightings
Skills Framework	TSC: Data Analytics (ATP-PIN-3001-1.1)
Mode	Instructor-led classroom facilitation with hands-on data labs in the browser (Killercoda Ubuntu, SQLite, Python/pandas) and browser-based analysis tools
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Integrate multiple datasets to extract and process data efficiently. (K4, A1)
- LO2: Conduct data mining to uncover and analyze trends using analysis techniques. (K1, A2)
- LO3: Perform statistical data analysis to derive actionable insights. (K2, A3)
- LO4: Present analytical outputs visually to communicate data effectively for decision-making. (K3, A5)
- LO5: Recognize and interpret sequential patterns to establish linkages between variables. (A4)

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, individual, summative, open book.
- Practical Performance (PP) — hands-on scenario-based data tasks, 1 hour, individual, summative, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 5 from 4:00 pm to 6:00 pm (WA 1 hour followed by PP 1 hour).
- A minimum of 75% attendance is required to be eligible for assessment and funding, and the learner must be assessed 'Competent' in both instruments.

Course Schedule

Day 1 — Data Concepts, Schemas and Environments

Time	Duration	Topic / Activity	Slides
9:00–9:30	30 min	Welcome, trainer and learner introductions, ground rules, learning outcomes, course outline and mandatory digital attendance (AM)	1–19
9:30–10:30	60 min	**Domain 1 — Data Concepts: data schemas, relational vs non-relational databases, the four NoSQL families, normalisation 1NF-5NF (concepts + demo)**	21–61
10:30–10:45	15 min	Tea break	—
10:45–13:00	135 min	Domain 1 — keys, relationships, referential integrity, OLTP vs OLAP. Hands-on: Lab 1 — Build a Relational Schema and Prove Referential Integrity	44–50
13:00–14:00	60 min	Lunch break	—
14:00–15:30	90 min	Domain 1 — warehouses, marts, lakes and lakehouses, star vs snowflake schemas, slowly changing dimensions, data types and structures. Hands-on: Lab 2 — Compare Structured, Semi-Structured and Unstructured Data	51–55
15:30–15:45	15 min	Tea break	—
15:45–17:45	120 min	Domain 1 — file formats, data languages, infrastructure (cloud/on-premise/containers), AI concepts and data sources. Hands-on: Lab 3 — Profile Machine/Log Data as a Data Source (PCAP Analyzer)	56–60
17:45–18:00	15 min	Day 1 recap, Q&A and PM digital attendance	1–19

Total training time: 480 minutes (8 hours).

Day 2 — Data Acquisition, Exploration and Preparation

Time	Duration	Topic / Activity	Slides
9:00–9:15	15 min	Day 1 recap and mandatory digital attendance (AM)	1–19
9:15–10:30	75 min	**Domain 2 — Data Acquisition: ETL vs ELT, full and delta loads, APIs (pull/push), web scraping, public repositories, surveys and sampling (concepts + demo)**	63–103
10:30–10:45	15 min	Tea break	—
10:45–13:00	135 min	Domain 2 — data profiling procedure, the defect classes: missing values, duplicates, redundancy, invalid data and outliers. Hands-on: Lab 4 — Explore a Dirty Dataset: Missing Values, Duplicates and Outliers	79–84
13:00–14:00	60 min	Lunch break	—
14:00–15:30	90 min	Domain 2 — data transformation: recoding, derived variables, imputation, aggregation, transposing, appending, parsing strings. Hands-on: Lab 5 — Build Parsing Patterns in RegexLab and Apply Them	85–90
15:30–15:45	15 min	Tea break	—
15:45–17:45	120 min	Domain 2 — text/date/logical functions, SQL joins, filtering, indexing, subqueries and query execution plans. Hands-on: Lab 6 — Derive Structured Fields from Raw Address Data (IP Calculator); Lab 7 — Integrate Multiple Datasets with SQL Joins and Cleanse the Result	91–102
17:45–18:00	15 min	Day 2 recap, Q&A and PM digital attendance	1–19

Total training time: 480 minutes (8 hours).

Day 3 — Data Analysis — Descriptive and Inferential Statistics

Time	Duration	Topic / Activity	Slides
9:00–9:15	15 min	Day 2 recap and mandatory digital attendance (AM)	1–19
9:15–10:30	75 min	**Domain 3 — analysis types (exploratory, performance, trend, gap, link) and descriptive statistics: mean, median, mode, range, variance, standard deviation (concepts + demo)**	105–141
10:30–10:45	15 min	Tea break	—
10:45–13:00	135 min	Domain 3 — z-scores, normal distribution, the empirical rule, parametric vs nonparametric data, frequency and histograms. Hands-on: Lab 8 — Descriptive Statistics and the Outlier That Moves the Mean	120–126
13:00–14:00	60 min	Lunch break	—
14:00–15:30	90 min	Domain 3 — inferential statistics: hypothesis testing, t-tests, p-values, Type I and Type II errors, chi-square, confidence intervals. Hands-on: Lab 9 — Hypothesis Testing: t-test, p-value and the Two Error Types	127–133
15:30–15:45	15 min	Tea break	—
15:45–17:45	120 min	Domain 3 — correlation, regression, R-squared, causation, troubleshooting analysis issues and communicating results to different audiences. Hands-on: Lab 10 — Correlation, Regression and the Causation Trap	134–140
17:45–18:00	15 min	Day 3 recap, Q&A and PM digital attendance	1–19

Total training time: 480 minutes (8 hours).

Day 4 — Visualization, Dashboards and Reporting

Time	Duration	Topic / Activity	Slides
9:00–9:15	15 min	Day 3 recap and	1–19

		mandatory digital attendance (AM)	
9:15–10:30	75 min	**Domain 4 — chart selection by question type: composition, comparison, distribution, relationship and trend; line, bar, pie, histogram, scatter (concepts + demo)**	143–170
10:30–10:45	15 min	Tea break	—
10:45–13:00	135 min	Domain 4 — specialist charts (tree map, heat map, bubble, waterfall, combination), geographic maps: dot, filled and layered. Hands-on: Lab 11 — Choose the Right Chart: Five Questions, Five Chart Types	155–161
13:00–14:00	60 min	Lunch break	—
14:00–15:30	90 min	Domain 4 — expressing business requirements, report and dashboard components, documentation elements, report types. Hands-on: Lab 12 — Build a KPI Dashboard and Validate Its Accuracy	162–169
15:30–15:45	15 min	Tea break	—
15:45–17:45	120 min	Domain 4 — dashboard design principles, filtering and sorting, drillthrough and tooltips, validating reporting accuracy, validation vs verification. Dashboard build review and critique	143–170
17:45–18:00	15 min	Day 4 recap, Q&A and PM digital attendance	1–19

Total training time: 480 minutes (8 hours).

Day 5 — Data Governance, Quality, Controls and Final Assessment

Time	Duration	Topic / Activity	Slides
9:00–9:15	15 min	Day 4 recap and mandatory digital attendance (AM)	1–19
9:15–10:30	75 min	**Domain 5 — data governance, the data lifecycle, data roles (owner, steward,**	172–203

		custodian, privacy officer), classification and compliance (concepts + demo)	
10:30–10:45	15 min	Tea break	—
10:45–12:15	90 min	Domain 5 — privacy and protection: access control, encryption at rest/in transit/in use, masking, de-identification and pseudonymisation. Hands-on: Lab 13 — Classify, Mask and De-Identify a Dataset (PDPA/GDPR)	186–191
12:15–13:00	45 min	Domain 5 — retention, preservation, destruction and sanitisation; regulations (PDPA, GDPR, HIPAA, SOX) and data sovereignty. Hands-on: Lab 14 — Assess Data-Leakage Risk and Set Access Controls	192–196
13:00–14:00	60 min	Lunch break	—
14:00–15:15	75 min	Domain 5 — documentation, versioning and data lineage; the six quality dimensions; profiling, monitoring and automated testing; master data management. Hands-on: Lab 15 — Automated Quality Assurance: Profile, Rule, Monitor	197–202
15:15–15:25	10 min	Tea break	—
15:25–15:45	20 min	Course revision across all five exam domains, CompTIA Data+ certification exam guidance and Q&A	172–203
15:45–15:55	10 min	Course feedback and mandatory TRAQOM survey on the LMS	1–19
15:55–16:00	5 min	**Briefing for Assessment and Assessment digital attendance**	196–200
16:00–17:00	60 min	**Written Assessment (WA) — Short-Answer Questions**	196–200

		(SAQ), 1 hour, individual, open book	
17:00–18:00	60 min	**Practical Performance (PP) — scenario-based hands-on data tasks, 1 hour, individual, open book. PM digital attendance**	196–200

Total training time: 480 minutes (8 hours).

Slide numbers refer to the Trainer Slides deck WSQ - CompTIA Certified Data+ Training-v5.0. Any change to the deck requires this column to be re-checked.

Lab Reference (aligned to the CompTIA Data+ exam domains)

Exam domain / Learning Unit	Exam weighting	Labs	Slides
Domain 01 (LU1 / LO1): Data Concepts and Environments	20%	Lab 1, Lab 2, Lab 3	21–61
Domain 02 (LU2 / LO2): Data Acquisition and Preparation	22%	Lab 4, Lab 5, Lab 6, Lab 7	63–103
Domain 03 (LU3 / LO3): Data Analysis	24%	Lab 8, Lab 9, Lab 10	105–141
Domain 04 (LU4 / LO4): Visualization and Reporting	20%	Lab 11, Lab 12	143–170
Domain 05 (LU5 / LO5): Data Governance, Quality and Controls	14%	Lab 13, Lab 14, Lab 15	172–203